

Do Thought Streams Matter?

Sharma, Nagaonkar, Choithani & Trivedi · [VideoDB](#) · arXiv 2604.11177 · April 2026
A complete deep-dive into every aspect of the paper

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01 · Motivation & Research Questions

Vision-language models (VLMs) like Gemini 2.5 now support **extended thinking**: before producing a final structured answer, the model generates an internal chain of thought — what this paper calls a **"thought stream."** Most benchmarks treat models as black boxes and evaluate only the final output. This paper does something different: it cracks open the reasoning trace and examines its relationship to the final answer.

The authors at VideoDB process large volumes of video content in production and need practical answers to three questions:

RESEARCH QUESTIONS

Q1: Does the thought stream actually contain useful scene information, or is it mostly filler and

meta-commentary?

Q2: Does the final output faithfully reflect what was reasoned about? Or does content get lost or fabricated?

Q3: Do different thinking budgets change what the model pays attention to? For instance, does a constrained model default to generic labels like "person" instead of specific identifications?

The key insight is shifting evaluation from *"how good is the output?"* to *"how does reasoning translate into the output?"* — measuring internal consistency rather than external correctness.

02 · Dataset & Scene Extraction

VIDEO HOURS

~100

VISUAL STYLES

37

CONTENT DOMAINS

38

TOTAL SCENE RESULTS

93,000+

Visual Styles (37 categories)

The dataset covers a wide spectrum: 2D and 3D animation, cinematic footage, documentary, gameplay and esports, live concert and event captures, motion graphics, surveillance footage, social media clips, vlogs, and vintage film. This diversity is deliberate — it tests whether the models can handle different visual idioms, not just well-lit cinematic shots.

Content Domains (38 categories)

Entertainment, sports, news, educational, culinary, music performance, drama, comedy, reality television, gaming, corporate, travel vlog, children's content, and many more. The breadth ensures the findings aren't specific to a single genre.

Production Quality Distribution

High Quality

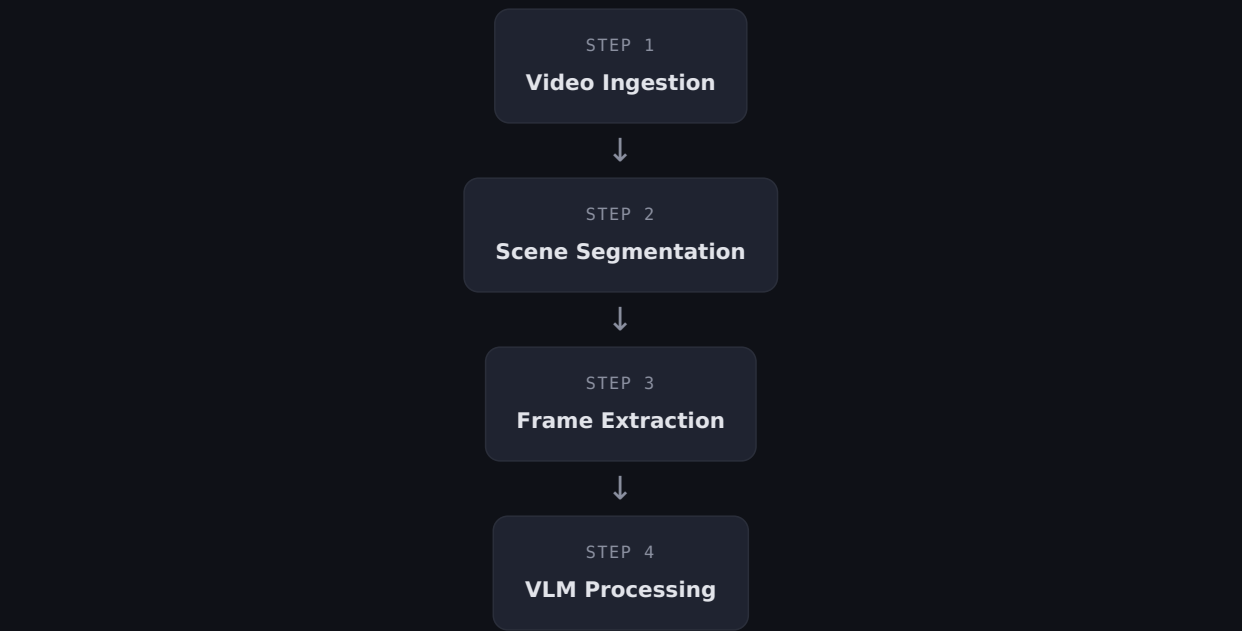
63.6%

Medium

34.1%

Weighted toward professionally produced content, which is realistic for production use cases but means low-quality video (e.g., surveillance, user-generated content) may be underrepresented.

Scene Extraction Pipeline



VideoDB segments videos into individual scenes using visual and semantic boundaries. Frames are extracted at **1 FPS** with a cap of **10 frames per scene**. Each scene is processed independently — there is no cross-scene context. This is an important design choice: it tests per-scene understanding only, not temporal reasoning across scenes or narrative comprehension.

The structured JSON output extracted per scene includes: subjects, actions, settings, emotions, shot types, and additional metadata.

03 · Model Configurations

Two model tiers of Gemini 2.5, tested with different reasoning budgets. All variants receive **identical prompts, schemas, and temperature settings**. The only independent variable is the thinking budget — how many tokens the model can spend on its thought stream.

VARIANT	TIER	BUDGET	MEAN THOUGHT TOKENS	MEAN TOTAL TOKENS
Flash - 128 Flash	Flash	128 tokens	105	~2,331
Flash - Dynamic Flash	Flash	Unlimited	1,021	~3,258
Lite - 512 Lite	Flash Lite	512 tokens	366	~2,563
Lite - 1024 Lite	Flash Lite	1,024 tokens	718	~2,919

DESIGN NOTE

Note that the actual thought tokens used are always less than the budget cap. Flash 128 averages

105 thought tokens (82% of budget), while Lite 512 averages 366 (71%). The model doesn't always use its full budget — it stops when it considers its reasoning sufficient.

04 · Token Anatomy

Each API call involves four categories of tokens. Understanding these is critical for interpreting the cost analysis.

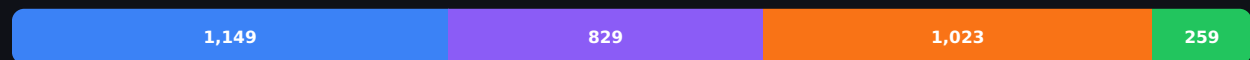
Token Breakdown Per Scene (Mean)

Each bar below shows the four components. The key variable is thought tokens (orange) — everything else is relatively stable.

Flash 128 — Total: 2,331



Flash Dynamic — Total: 3,260



Lite 512 — Total: 2,564



Lite 1024 — Total: 2,919



● Image Tokens (input) ● Text Tokens (input) ● Thought Tokens (output) ● Response Tokens (output)

KEY OBSERVATIONS

Input tokens (~1,964-1,978) consist of text prompt + image frames. They vary per scene (depending on frame count) but stay consistent across variants for the same scene, since every variant sees the same frames.

Response tokens (~222-262) are remarkably stable across all variants. The final structured JSON output is roughly the same length regardless of how much the model thinks.

Thought tokens (105-1,021) are where all the cost variation lies. This is the sole independent variable.

05 · The Three Metrics — Deep Dive

This is the paper's central contribution: three metrics that measure different dimensions of the relationship between thinking and output. Existing benchmarks only evaluate the final answer. These metrics let you look *inside* the process.

METRIC 1

Contentfulness

How much of the thought stream is actual scene content vs. filler?

METRIC 2

Thought-Final Coverage

How faithfully does the thought stream translate into the final output?

METRIC 3

Dominant Entity Analysis

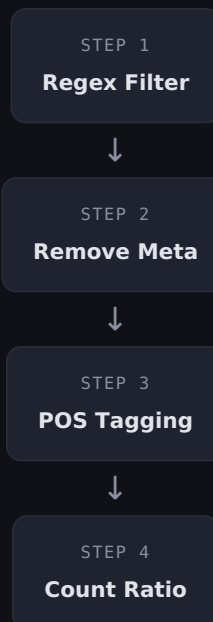
Which subjects, actions, and settings does each variant focus on?

Crucially, **GPT-5** serves as an independent judge for the coverage metrics. Contentfulness is fully deterministic (regex + POS tagging). The use of a different model family (OpenAI) as judge avoids the circularity of having Gemini evaluate itself.

06 · Contentfulness in Detail

Contentfulness $\in [0, 1]$ measures what fraction of the thought stream consists of actual scene-related content (nouns and verbs describing what's in the video) versus **meta-commentary** (the model narrating its own reasoning process).

How It's Computed



Step 1-2: Sentences containing meta-commentary are identified using regex patterns that match phrases like "I will," "let me," "step by step," "json," "I need to think about," etc. These sentences are removed entirely.

Step 3: The remaining sentences are POS-tagged using NLTK. Only words belonging to **noun phrases** (the scene's subjects and objects) or **verb phrases** (the actions) are counted as content words.

Step 4: Contentfulness = content words / total words in the full thought stream.

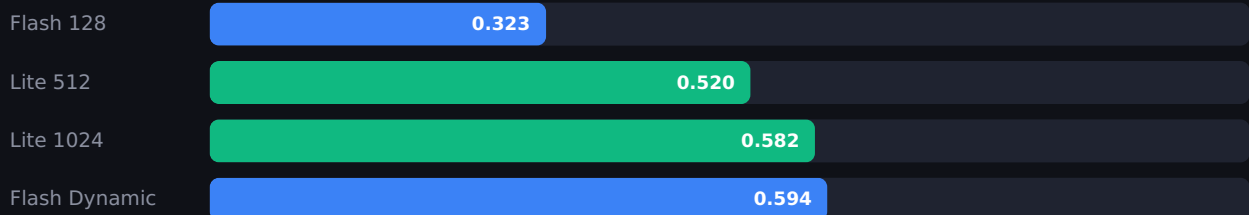
WORKED EXAMPLE

"Let me analyze this scene carefully." "A young woman sits at a wooden desk, typing on a silver laptop in a bright office."

The first sentence is meta-commentary → removed. Content words from the remaining text: woman , sits , desk , typing , laptop , office = 6 content words. Full trace has 20 total words.

$$\text{Contentfulness} = 6 / 20 = 0.30$$

Scores Across Variants



INSIGHT

Contentfulness rises roughly linearly with the thinking budget — unlike F1 which plateaus. This makes sense: with more tokens available, the model has room for both meta-commentary AND scene content. But the meta-commentary doesn't go away entirely even with unlimited budgets — Flash Dynamic's 0.594 means ~40% of its thought stream is still process narration.

This is where Flash and Lite diverge stylistically. At the same contentfulness level, Lite achieves it with fewer total tokens because it skips more of the "let me think about this" narration.

07 · Thought-Final Coverage — Deep Dive

This is the most complex and arguably most important metric. It measures how the thought stream and final output align, using GPT-5 as an extraction judge.

The Two Sub-Metrics

Thought Coverage (TC)

"Of everything the model *thought about*, how much made it into the final output?"

Low TC → model reasoned about details then **dropped** them

Output Grounding (OG)

"Of everything in the *final output*, how much was actually present in the thought stream?"

Low OG → model **added** content it never reasoned about

$$F1 = 2 \times TC \times OG / (TC + OG)$$

How the Matching Works

The process uses **cascaded fuzzy matching** with three tiers:

Tier 1 — Exact match: Straightforward string equality.

Tier 2 — Token-sort ratio ≥ 75 : Both strings are tokenized, the tokens are sorted alphabetically, then similarity is computed. This catches reorderings like "wooden desk" matching "desk, wooden."

Tier 3 — Partial ratio ≥ 75 : Finds the best matching substring of the shorter string within the longer one. Catches cases like "laptop" matching "silver laptop."

This cascaded approach is sensible: exact matches are checked first (cheapest), then increasingly fuzzy methods are tried only when needed. The 75% threshold for both fuzzy methods is a judgment call — the paper doesn't ablate this value.

WORKED EXAMPLE

Thought stream items: [woman, desk, laptop, office, typing, focused expression]

Final output items: [woman, desk, laptop, office, typing, **smiling**]

5 of 6 thought items appear in output → **TC = 5/6 $\approx$ 0.833**

5 of 6 output items were in thought stream ("smiling" was not) → **OG = 5/6 $\approx$ 0.833**

$$F1 = 2 \times (5/6 \times 5/6) / (5/6 + 5/6) \approx 0.833$$

"Smiling" appearing in the output without being in the thought stream is a case of compression-step hallucination.

What GPT-5 Does as Judge

GPT-5 performs the extraction step: it reads the raw thought stream text and the structured JSON output, then extracts "atomic items" — individual facts like a specific subject, action, or attribute. These atomic items are what get matched. The actual matching (the fuzzy comparison) is deterministic, but the extraction of atomic items from messy text requires judgment, which is why an LLM judge is used.

CRITICAL CAVEAT

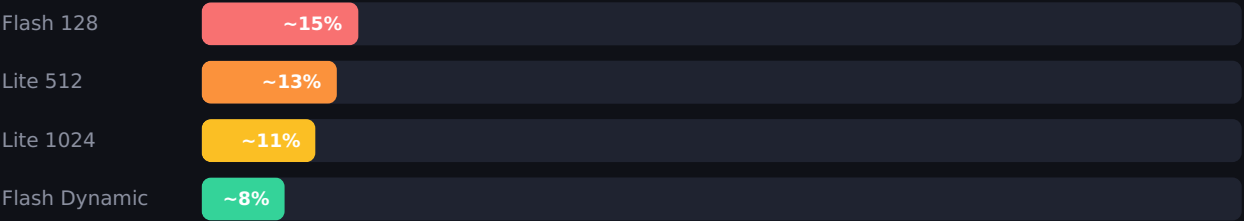
These metrics measure **internal consistency** between the thought stream and the output, NOT correctness against external ground truth. A model could achieve perfect TC, perfect OG, and perfect F1 while being completely wrong about what's in the scene. The model could "think" about a cat and "output" a cat while the video actually shows a dog. This framework can't detect that.

08 · Dominant Entity Analysis

For each scene, the authors identify the single most prominent **subject**, **action**, and **setting**. This lets them compare attentional focus across variants and answer the question: does a constrained model default to vague labels?

Subject Specificity: "person" as a Canary

The clearest signal is how often models default to the generic label "person" instead of identifying specific subjects:



"% of scenes where 'person' is the dominant subject" — lower is better (more specific identification)

With more thinking budget, models are more likely to identify specific subjects — "streamer," "chef," "cat" — rather than falling back on safe, generic terms. This has direct production implications: if you're building a search index over video content, the specificity of entity labels determines how useful the index is.

09 · Full Results Breakdown

VARIANT		SCENES	CONTENT.	T.COVS	O.GRD.	F1	ERR.%
Flash	Flash 128	25,807	0.323	0.853	0.767	0.830	0.14%
Flash	Flash Dynamic	22,519	0.594	0.953	0.964	0.957	0.13%
Lite	Lite 512	23,091	0.520	0.940	0.948	0.942	0.09%
Lite	Lite 1024	22,188	0.582	0.954	0.966	0.959	0.12%

Reading the Results

Flash 128 — The Budget-Constrained Baseline

The most scenes processed (25,807), but the lowest quality across every metric. The critical number is **Output Grounding at 0.767** — meaning roughly 1 in 4 items in the final JSON were never mentioned in the thought stream. The model was forced to "wing it" when compressing into the final answer. Contentfulness is also the lowest at 0.323 — two-thirds of the thought stream is meta-commentary rather than scene content.

Flash Dynamic — Maximum Thinking

With unlimited budget, Flash produces the most contentful thought streams (0.594) and very high alignment scores. But it's NOT the top performer on F1 — Lite 1024 beats it by a hair (0.959 vs 0.957) while using 30% fewer thought tokens.

Lite 512 — The Efficiency Sweet Spot

At ~2,563 total tokens per scene (only 10% more than Flash 128), Lite 512 achieves $F1 = 0.942$ — a massive improvement over Flash 128's 0.830. For most production use cases, this might be the optimal configuration: it costs far less than the higher-budget variants while delivering quality that's close to the ceiling.

Lite 1024 — The Quality Leader

Highest F1 (0.959), highest Thought Coverage (0.954), highest Output Grounding (0.966). It spends 718 thought tokens on average — far less than Flash Dynamic's 1,021 — but achieves better results. This is the paper's strongest evidence that Flash Lite is simply more token-efficient for this task.

SCENE COUNT VARIATION

Note that scene counts differ across variants (22,188 to 25,807). The error rates are small (0.09–0.14%), but Flash 128's higher scene count likely reflects fewer processing failures due to its cheaper cost. The scene count differences should be accounted for when comparing, though at these volumes the means are statistically robust.

10 • Scaling & Diminishing Returns

This is one of the paper's most practically important findings. The two left-panel metrics from Figure 2 tell different stories:

Contentfulness: Linear Scaling

Contentfulness rises roughly linearly as you add more thinking tokens: $0.323 \rightarrow 0.520 \rightarrow 0.582 \rightarrow 0.594$. More tokens = proportionally more scene content. This makes intuitive sense — with more room, the model can describe more of what it sees.

F1: Steep Then Flat

F1 shows a very different curve. The jump from 105 tokens to 366 tokens yields a +0.112 gain ($0.830 \rightarrow 0.942$). But the next jump, from 366 to 718 tokens (nearly double), gains only +0.017 ($0.942 \rightarrow 0.959$). And going all the way to 1,021 tokens actually *loses* 0.002 ($0.959 \rightarrow 0.957$).

PLATEAU ZONE

Beyond ~400 thought tokens, additional thinking produces negligible F1 improvement for this task.

The "knee" of the curve sits between 100 and 400 tokens. Below 100, the model is severely constrained and hallucinates at the compression step. Above 400, you're paying for contentfulness (the model thinks about more stuff) but without meaningful improvement in the thought-to-output alignment that F1 measures.

The Implication

For production use, this suggests a clear budget strategy: allocate enough thinking tokens to get past the knee of the curve (somewhere around 300–500 tokens for scene understanding), then stop. Additional tokens buy you richer internal traces but don't meaningfully improve the quality of the final output.

The authors are careful to note this pattern is specific to their dataset and task — longer-horizon video reasoning or more complex extraction tasks might shift the plateau point.

11 · Consistency & Quality Tiers

Averages can hide important variation. Table 3 breaks down the *distribution* of quality — how consistent each variant is across scenes.

VARIANT	F1 MEAN	F1 STD DEV	CV (Σ/M)	PERFECT (F1=1.0)	LOW (F1 < 0.5)
Flash 128	0.830	0.234	0.282	36.5%	11.0%
Flash Dynamic	0.957	0.079	0.082	62.0%	0.3%
Lite 512	0.942	0.097	0.103	55.8%	0.6%
Lite 1024	0.959	0.078	0.082	64.3%	0.2%

What the CV Reveals

The **coefficient of variation** ($CV = \sigma/\mu$) measures relative variability. Flash 128's CV of 0.282 is dramatically higher than the other variants (~0.08-0.10). This means Flash 128 isn't just worse on average — it's *unpredictable*. Some scenes work fine (36.5% achieve perfect scores), but 11% of scenes fall below 0.5, which represents a severe failure.

By contrast, Lite 1024 achieves perfect alignment on 64.3% of scenes and has a catastrophic failure rate of just 0.2%. In production, this consistency matters as much as the average score.

THE FLASH 128 PROBLEM

Flash 128's bimodal distribution is the worst of both worlds: it sometimes works perfectly (36.5%), giving you confidence, but it fails badly often enough (11.0%) to be unreliable. If you deployed this and spot-checked a handful of results, you might think it's fine — but 1 in 9 scenes would have significantly degraded alignment between reasoning and output.

12 · Thought Stream Similarity

Perhaps the paper's most surprising finding: despite being different model tiers, Flash and Flash Lite produce remarkably similar thought streams.

COMPARISON	VARIANT A	VARIANT B	SIMILARITY
Cross-tier (Flash ↔ Lite)	Lite 1024	Flash Dynamic	0.887
	Lite 512	Flash Dynamic	0.881
	Lite 1024	Flash 128	0.885
	Lite 512	Flash 128	0.880

Within-tier	Lite 1024	Lite 512	0.905
	Flash 128	Flash Dynamic	0.903
Determinism Test	Flash Dyn.	Flash Dyn. (rerun)	0.893

Three Takeaways

Cross-tier similarity (~0.88) is nearly as high as within-tier (~0.90). Flash and Lite think about the same things. The difference between "same model, different budget" and "different model, different budget" is only about 2 percentage points of similarity. This suggests the two tiers share underlying reasoning patterns — possibly similar training data, architecture, or fine-tuning.

The determinism test is revealing. When Flash Dynamic was run twice on the same videos, similarity was only 0.893 — meaning even the same model on the same input doesn't produce identical thought streams. The fact that cross-tier similarity (0.887) is within 1% of same-model determinism (0.893) is striking. Areas that remain stable across runs include brand names, logos, text detection, and object identity. Areas that vary include emotion labels and location descriptions — these are inherently more subjective.

The style difference matters. Flash tends to narrate its reasoning process ("Let me think about what's in this image..."), while Lite jumps straight to describing the scene. They think about the same entities, but Lite wastes fewer tokens on process talk. This is why Lite achieves comparable contentfulness with fewer tokens.

13 • Compression-Step Hallucination

This is one of the paper's most important conceptual contributions: a new category of model failure that sits between traditional hallucination and reasoning failure.

DEFINITION

Compression-step hallucination occurs when the final structured output includes information that was NOT explicitly present in the model's verbalized thought stream. It's a mismatch between the reasoning trace and the output — the model "adds" content during the compression step from thinking to structured JSON.

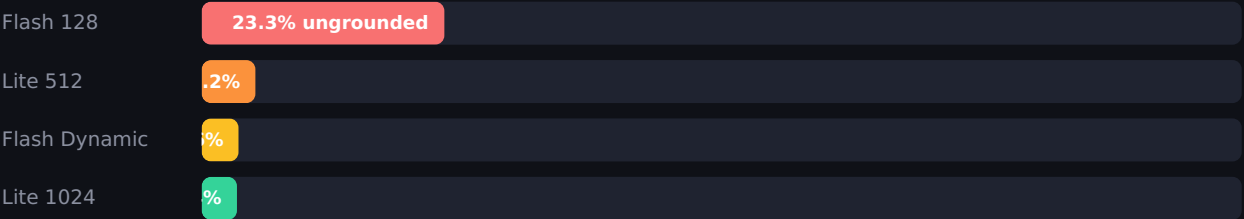
Important Nuance

The authors are careful to note this does NOT necessarily mean the model didn't internally consider the information. It could have processed it in ways not captured by the thought stream (which is only an "observable trace," not a complete record of internal computation). The mismatch might reflect a limitation of the thought stream as a window into the model's reasoning, not necessarily a true hallucination in the sense of fabrication.

But from a practical standpoint, it's a red flag: if you can't find the basis for an output claim in the reasoning trace, you have no way to audit or verify it.

The Evidence

Flash 128's Output Grounding is 0.767, meaning ~23.3% of output items had no basis in the thought stream. Compare this to Lite 1024 at 0.966 (only ~3.4% ungrounded). The mechanism is straightforward: with a tight token budget, the model can't build a sufficiently detailed reasoning trace. When it then generates the structured output, it has to fill in gaps — and those gap-fills are compression-step hallucinations.



Percentage of output items with no match in the thought stream (1 - Output Grounding)

Why This Matters

Traditional hallucination research focuses on models saying factually wrong things. Compression-step hallucination is different — it's about the model's output diverging from its own reasoning, regardless of factual accuracy. This is especially concerning for applications where auditability matters: if you need to explain *why* the model made a particular metadata extraction, and the reasoning trace doesn't support it, you have an audit gap.

14 · Flash vs. Flash Lite — Head to Head

The paper's Figure 3 provides a direct side-by-side comparison. Within each tier, higher budgets improve all metrics. But the more interesting comparison is *across* tiers.

Gemini 2.5 Flash

Flash 128: Cont. **0.324** | TC 0.874 | OG 0.813 | F1 0.830

Flash Dyn.: Cont. **0.595** | TC 0.953 | OG 0.966 | F1 0.957

Gemini 2.5 Flash Lite

Lite 512: Cont. **0.521** | TC 0.941 | OG 0.949 | F1 0.942

Lite 1024: Cont. **0.582** | TC 0.954 | OG 0.968 | F1 0.959

The Token Efficiency Gap

Lite 1024 matches or exceeds Flash Dynamic on every metric while using **30% fewer thought tokens** (718 vs. 1,021). The savings come from two places: fewer thought tokens and slightly shorter final responses (225 vs. 259 response tokens).

The paper's explanation: Lite spends more of its budget on **scene content** and less on **process narration**. When Flash writes "Let me examine this frame carefully. I can see that there appears to be a person..." Lite writes "A person sits at a desk..." The same information reaches the final output, but Lite gets there with fewer tokens.

For video scene understanding at scale, Flash Lite with a 1024-token budget appears to be the dominant strategy in this evaluation. It costs less per scene than Flash Dynamic, produces equal or better quality, and achieves the highest consistency (lowest CV, highest perfect-score rate). The only scenario where Flash Dynamic might be preferred is if you need maximum contentfulness in the thought stream itself (e.g., for debugging or audit trails).

15 · Five Key Takeaways

1 More thinking helps, but gains plateau quickly.

Most quality improvement happens in the first few hundred thought tokens. Beyond ~700 tokens, additional thinking adds cost with diminishing returns for this task. The "knee" of the F1 curve is between 100–400 tokens.

2 Lite 1024 is the quality leader.

Best F1 (0.959), best Thought Coverage (0.954), best Output Grounding (0.966), highest perfect-score rate (64.3%), and lowest catastrophic failure rate (0.2%) — all while using 30% fewer thought tokens than Flash Dynamic.

3 Tight budgets increase compression-step hallucination.

Flash 128 outputs details that were never explicitly present in its thought stream in ~23% of items. This is a qualitatively different failure mode from traditional hallucination — it's a mismatch between the model's own reasoning trace and its output.

4 Flash and Lite think about the same things.

Cross-tier thought stream similarity (~0.88) is nearly as high as same-model determinism (~0.89), suggesting the two tiers share underlying reasoning patterns. They differ in style (Flash narrates its process; Lite describes the scene), not in substance.

5 Flash Lite is more token-efficient.

It spends less on process narration and more on scene content, achieving comparable or better quality at lower cost. For production video understanding, it's the better choice in this evaluation.

16 · Limitations & Future Work

Acknowledged Limitations

LIMITATION 1 – CONSISTENCY ≠ CORRECTNESS

The metrics measure thought-to-output alignment, NOT whether the model correctly identified what's in the video. A model with high F1 could still be wrong — it just means it was *consistently* wrong between its thinking and output. Pairing these metrics with human ground truth is explicitly called out as the most important next step.

LIMITATION 2 – SINGLE LLM JUDGE

Coverage metrics depend on GPT-5 for atomic item extraction. A single judge model could introduce systematic biases — perhaps it consistently extracts certain types of items better than others, or its extraction patterns favor one model variant over another. Multi-judge validation or human calibration would strengthen the results.

LIMITATION 3 – SCENE-LEVEL ONLY

The dataset processes scenes independently at 1 FPS with ≤ 10 frames per scene. This doesn't test: long-range temporal reasoning, multi-scene dependencies, narrative structure understanding, fast-action sequences (1 FPS is too slow), or scenes requiring context from surrounding content.

LIMITATION 4 – SINGLE MODEL FAMILY

Only Gemini 2.5 variants are tested. The findings may not generalize to other VLMs (GPT, Claude, open-source models). The thought stream behavior, the diminishing returns curve, and the cross-tier similarity could all be Gemini-specific.

Planned Future Work

The authors outline several directions: pairing internal consistency metrics with human ground truth, extending to other VLM families, conducting a fine-grained budget sweep (64 to 2048 tokens), adding latency and dollar-cost profiling, domain-specific evaluation (medical, sports, surveillance), and larger-scale determinism testing.

17 · Related Work & Context

The paper positions itself at the intersection of several research threads:

Model Benchmarking

MMLU (Hendrycks et al., 2021): 57-subject multiple-choice benchmark for broad knowledge. **HellaSwag** (Zellers et al., 2019): commonsense reasoning via plausible continuations with adversarial distractors. **HumanEval** (Chen et al., 2021): functional code generation evaluated against unit tests. All these evaluate final outputs only — this paper adds the "look inside" dimension.

Chain-of-Thought Research

Wei et al. (2022) showed that adding "let's think step by step" to prompts significantly improves reasoning performance, particularly in large models. This paper builds on that by asking: if the model is thinking step by step, how *good* is that thinking? How much of it is useful content vs. filler? How much of it survives into the final answer?

Video Understanding Benchmarks

ActivityNet (2015): 20K YouTube videos, 200 activity classes. **Ego4D** (2022): 3,670 hours of first-person video for episodic memory and forecasting. **Video-MME** (2024): first comprehensive multi-modal LLM

video benchmark. This paper's task (structured metadata extraction) is closer to production needs than academic classification tasks.

LLM-as-Judge

Zheng et al. (2023) introduced MT-Bench and Chatbot Arena, demonstrating that strong LLMs can serve as reliable evaluators with >80% agreement with human annotators. This paper uses GPT-5 as judge, which is a natural extension — though using a single judge model is a recognized limitation.

18 · Production Implications

While the paper is positioned as a research contribution, it comes from a production video processing company (VideoDB), and several findings have direct practical implications.

Budget Allocation Strategy

RECOMMENDATION FROM THE DATA

For video scene understanding at scale: Use Flash Lite with a 512-1024 token thinking budget. The quality difference between 512 and 1024 is small (~ 0.017 F1), so the choice depends on your cost sensitivity. Below 300 tokens, you're in the danger zone for compression-step hallucination.

Avoid: Flash with tight budgets (128 tokens). The quality degradation is severe, and the high variance means you can't reliably predict which scenes will fail.

Auditability

If you need to audit *why* the model made a particular metadata extraction (for compliance, quality assurance, or debugging), the thought stream is your audit trail. But with tight budgets, $\sim 23\%$ of output items have no basis in that trail. Higher budgets reduce this gap to $\sim 3\%$, making the thought stream a more reliable audit artifact.

Search Index Quality

The dominant entity analysis shows that constrained models produce more generic labels. If you're building a video search index, the difference between "person" and "chef" is the difference between a useless and a useful search result. Budget allocation directly affects the specificity and therefore the utility of your metadata.

Cost Modeling

At $\sim 2,563$ tokens per scene (Lite 512) vs. $\sim 3,260$ tokens per scene (Flash Dynamic), the difference is $\sim 27\%$ more tokens for Flash Dynamic — with arguably worse results. Across 93,000+ scenes, this adds up to millions of tokens of unnecessary spend. The paper's data suggests the cheaper option is also the better one for this specific task.

The Broader Question

This paper's framework — measuring what's inside the reasoning trace, not just the final answer — is applicable beyond video understanding. Any task where models use extended thinking could benefit from

these metrics. The contentfulness measure could help identify models or configurations that waste tokens on self-narration. The coverage metrics could detect compression-step hallucination in any domain where reasoning traces are available.